

Customer Churn Analysis Report

Objective

The primary objective of this analysis is to identify and understand the key factors contributing to customer churn in a telecommunications company. By analyzing patterns across demographics, service usage, contract types, and payment methods, this report aims to uncover actionable insights that can inform targeted retention strategies, reduce churn rates, and improve overall customer lifetime value.

Overview

Customer churn represents a critical business challenge, particularly in the telecommunications industry, where retaining customers is often more cost-effective than acquiring new ones. This analysis explores the factors influencing customer churn using a dataset of **7,043 customers**, examining demographic, behavioral, and service-related features to identify churn drivers and provide actionable insights.

Dataset Summary

- **Total records:** 7,043 customers
- **Target variable:** Churn (Yes/No)
- **Features include:**
 - Customer demographics (e.g., SeniorCitizen, tenure)
 - Account details (e.g., Contract type, PaymentMethod)
 - Services subscribed (e.g., InternetService, StreamingTV, TechSupport)

Key Insights

1. Demographics & Tenure

- **Senior Citizens** make up about **16%** of the customer base. However, their churn rate is notably higher at approximately **42%**, compared to **26%** for non-senior citizens. This suggests that older customers are more likely to discontinue services.
- **Tenure** is inversely related to churn:
 - Customers with **less than 12 months** of tenure have a churn rate of **~45%**.

- Those with **more than 60 months** of tenure show a churn rate below **10%**.
- This trend emphasizes the importance of early engagement and retention efforts.

2. Contract Type & Customer Commitment

- Churn rates vary significantly by **contract type**:
 - **Month-to-Month: 43% churn rate**
 - **One-Year: 11% churn rate**
 - **Two-Year: 3% churn rate**
- Longer contracts are highly correlated with lower churn, highlighting the value of promoting annual or biennial plans to improve retention.

3. Service Subscription Patterns

Customers with limited or no value-added services are more prone to churn:

- **OnlineSecurity, TechSupport, and DeviceProtection** significantly reduce churn:
 - For instance, churn among those **without TechSupport** is **~30%**, while it's **~13%** for those with support.
 - Similarly, absence of **OnlineBackup** and **OnlineSecurity** correlates with higher churn.
- **Streaming Services** (TV and Movies):
 - Customers who use these services alone show moderate churn, but when **bundled with support features**, churn drops considerably.
- Customers with **No InternetService** (typically inactive or minimal-use customers) have very low churn, but they're not high-value from a revenue standpoint.

4. Internet & Phone Service Influence

- **Fiber optic** customers exhibit **higher churn** than **DSL** users, potentially due to pricing or service quality issues.
- **PhoneService** alone does not heavily influence churn, but combinations with **MultipleLines** and other bundled features show more stable behavior.

5. Payment Method & Billing Preferences

- **Electronic check** users churn at a rate of **~45%**, compared to only **~15%** for those using **Bank Transfers** or **Credit Cards**. This could point to trust issues or lack of automation in check processing.
- Customers using **Paperless Billing** are more likely to churn, possibly due to reduced reminders or engagement touchpoints.

Visual Insights

Visualizations in the analysis included:

- **Count plots** comparing churn across service categories.
- **Stacked bar plots** showing percentage breakdowns of churn within each category (e.g., by contract type, senior citizen status).
- **Tenure histograms** with churn overlay to depict retention over time.

These visuals made it clear that:

- Churn is clustered among short-tenure, month-to-month customers.
- Service engagement and bundling directly correlate with lower churn rates.

Strategic Recommendations

1. Targeted Retention Programs

- Develop onboarding campaigns and loyalty rewards for customers in their first year.
- Use predictive churn models to identify at-risk short-tenure customers.

2. Contract Optimization

- Provide incentives (discounts, loyalty points, premium features) to shift customers from **month-to-month** to **yearly contracts**.

3. Service Bundling

- Cross-sell value-added services like **OnlineSecurity** and **TechSupport**, especially to customers with only core services.

4. **Improve Customer Experience for Fiber Users**

- Investigate causes of dissatisfaction among **Fiber Optic** customers through surveys and customer feedback.

5. **Billing & Payment Improvements**

- Educate customers about the ease and reliability of **automated bank and card payments**.
- Consider personalized alerts and engagement campaigns for **paperless billing users**.

Conclusion

This analysis reveals that **customer tenure, contract type, service bundling, and payment preferences** are strong predictors of churn. By addressing these areas with focused strategies—particularly during the critical early months of a customer’s lifecycle—telecom providers can significantly reduce churn, increase customer satisfaction, and enhance long-term profitability.